

Chapter 1

Units and dimensions

Units

All physical quantities are measured in terms of a standard unit and all formulae will only work if a system of units is used. The units used by scientists and engineers are those known as the International System of Units or S.I. Some of the fundamental quantities of the system are shown in Table 1. Other quantities used in this book can be expressed in units derived from these.

Table 1

Quantity	Symbol for Quantity	Unit	Symbol for Unit
length	l	metre	m
mass	m	kilogram	kg
time	t	second	s
electric current	I	ampere	A
temperature	T	kelvin	K

For quantities much larger or smaller than a standard unit a system of prefixes is used. These are shown in Table 2.

Table 2

Prefix	Symbol	Factor
pico	p	10^{-12}
nano	n	10^{-9}
micro	μ	10^{-6}
milli	m	10^{-3}
kilo	k	10^3
mega	M	10^6
giga	G	10^9

There are a few units and prefixes which are in common use but are not part of the S.I. system. These are shown in Table 3.

Table 3

Unit	Symbol	S.I. Equivalent
centimetre	cm	10^{-2} m
minute	min.	60 s
hour	h	3.6×10^3 s
electron volt	eV	1.60×10^{-19} J.
atomic mass unit	u	1.66×10^{-27} kg
litre	L	10^{-3} m ³
tonne	—	10^3 kg

When using formulae, S.I. units should be used almost without exception.

Example 1

How far does a car travel in 6.0 min. if it is moving at 36 km h⁻¹?

$$\begin{aligned}
 t &= 6.0 \text{ min.} & v &= \frac{36 \times 1000}{3600} \text{ m s}^{-1} & t &= 6.0 \times 60 \text{ s} \\
 v &= 36 \text{ km h}^{-1} & s &= vt = 1.0 \times 10^1 \times 360 \\
 v &= \frac{s}{t} & s &= 3600 \text{ m}
 \end{aligned}$$

Distance travelled is $3.6 \times 10^3 \text{ m}$ or 3.6 km.

Dimensions

Most quantities used in physics are measured in derived units which are based on the fundamental units of mass, length, time, current and temperature. In mechanics, only mass, length and time are needed. The dimensions of any quantity show the relationship of that quantity to the fundamental dimensions of mass, M, length, L, and time, T. For example, velocity is displacement divided by time. Therefore

$$v = \frac{s}{t} = \frac{L}{T} = LT^{-1}$$

Similarly, force is mass times acceleration

$$F = ma = \frac{ML}{T^2} = MLT^{-2}$$

Square brackets can be used to mean 'the dimensions of'. Dimensions are useful to check formulae and can be used to derive formulae, since the dimensions of the quantities on each side of the equals sign must be the same.

Example 2

What are the dimensions of energy?

$$\begin{aligned}
 E_k &= \frac{1}{2}mv^2 \\
 [E_k] &= ML^2T^{-2}
 \end{aligned}$$

The dimensions of energy are ML^2T^{-2} .

Example 3

Show that the equation $s = ut + \frac{1}{2}at^2$ is dimensionally consistent.

$$\begin{aligned}
 [L.H.S.] &= L \\
 [R.H.S.] &= LT^{-1}T + LT^{-2}T^2 \\
 &= L = [L.H.S.]
 \end{aligned}$$

Hence the equation is dimensionally consistent.

Set 1

1 What are the S.I. units of:

- | | | |
|------------|----------|--------------|
| (a) length | (b) mass | (c) force |
| (d) energy | (e) time | (f) velocity |

2 Convert the following to S.I. units.

- | | | |
|-----------|----------------------------|----------------|
| (a) 2.5 g | (b) 3.6 km h ⁻¹ | (c) 5.0 tonnes |
|-----------|----------------------------|----------------|

3 Calculate the speed in m s⁻¹ of a car that travels 2.0 km in 1.0 min.

4 Calculate the distance travelled by a train moving at a constant speed of 72 km h⁻¹ in 15 min.

5 Express the following in S.I. units.

- | | | |
|-----------------------|------------|--------------|
| (a) 4.0 μA | (b) 6.5 pC | (c) 1.25 kPa |
| (d) 2.7 cm | (e) 5.0 Gm | (f) 1.11 MJ |

6 Which of the following are measured in fundamental units.

- | | | |
|-----------------|------------|---------------------|
| (a) force | (b) mass | (c) time |
| (d) temperature | (e) energy | (f) electric charge |

7 Convert the following to S.I. units.

- | | | |
|-----------|-------------------------|-------------------------|
| (a) 5.2 L | (b) 2.3 cm ³ | (c) 3.5 mm ² |
|-----------|-------------------------|-------------------------|

8 What are the dimensions of:

- | |
|--|
| (a) momentum (mass $\times$ velocity) |
| (b) work (force $\times$ displacement) |
| (c) impulse (force $\times$ time) |

9 Show that the equation for the final velocity of an object undergoing uniform acceleration is dimensionally consistent.

$$v^2 = u^2 + 2as$$

10 What are the dimensions of viscosity η if the equation for the force on a sphere moving through a fluid is $F = 6\pi\eta rv$ where r is the radius of the sphere, v is the speed and F is the force?

Answers

- | | | |
|--|--------------------------------------|--|
| 1 (a) metre (m) | (b) kilogram (kg) | (c) newton (N) |
| (d) joule (J) | (e) second (s) | (f) metres per second (m s ⁻¹) |
| 2 (a) $2.5 \times 10^{-3} \text{ kg}$ | (b) 1.0 m s^{-1} | (c) $5.0 \times 10^3 \text{ kg}$ |
| 3 33 m s^{-1} | 4 $1.8 \times 10^4 \text{ m}$ | |
| 5 (a) $4.0 \times 10^{-6} \text{ A}$ | (b) $6.5 \times 10^{-12} \text{ C}$ | (c) $1.25 \times 10^3 \text{ Pa}$ |
| (d) $2.7 \times 10^{-2} \text{ m}$ | (e) $5.0 \times 10^9 \text{ m}$ | (f) $1.11 \times 10^6 \text{ J}$ |
| 6 (b), (c) and (d) | | |
| 7 (a) $5.2 \times 10^{-3} \text{ m}^3$ | (b) $2.3 \times 10^{-6} \text{ m}^3$ | (c) $3.5 \times 10^{-6} \text{ m}^2$ |
| 8 (a) MLT^{-1} | (b) ML^2T^{-2} | (c) MLT^{-1} |
| 9 L^2T^{-2} | 10 $ML^{-1}T^{-1}$ | |

Chapter 2

Significant figures and standard notation

Significant figures

Significant figures are those digits that are known with certainty plus the first digit that is uncertain. When using values in calculations, the uncertainty in the last digit is carried through the working and appears in the solution. The various mathematical operations can result in uncertainty appearing at different places in the solution. When writing numbers with zeroes, problems can arise with interpretation, and certain rules must be followed. In this book the following apply.

- 1 All non-zero digits are significant. So 11.2 has three significant figures.
- 2 All zeroes between two non-zero digits are significant. So 1007 has four significant figures.
- 3 All zeros to the right of a decimal point and following a non-zero digit are significant. So 0.0120 has three significant figures.
- 4 All other zeroes are not significant. So in 500, only the 5 is significant. In some instances the zeroes may be significant, in which case there are three significant figures. In order to show this so there is no disagreement it must be written in standard notation, that is as 5.00×10^2 . Another example of this last rule is 0.0013. It has only two significant figures.

Standard notation

To avoid problems with ambiguity all quantities are best written as a number equal to or greater than one and less than ten multiplied by a power of ten.

Example 1

123 kg is written as 1.23×10^2 kg.

Example 2

120 kg is written as 1.2×10^2 kg.

Or in this problem, if the zero was significant, then

120 kg is written as 1.20×10^2 kg.

Example 3

0.0120 kg is written as 1.20×10^{-2} kg.

Exponential notation

When multiplying exponents of ten, add the exponent, and when dividing, subtract the exponent.

Example 4

Find the value of $10^{-6} \times 10^{+9}$.

$$10^{-6} \times 10^{+9} \qquad 10^{(-6+9)} = 10^3$$

The value is 10^3 .

Example 5

Find the value of $\frac{10^{-6}}{10^{+9}}$.

$$\frac{10^{-6}}{10^{+9}} \qquad 10^{(-6-9)} = 10^{-15}$$

The value is 10^{-15} .

When adding or subtracting quantities expressed in standard notation, the exponents must be the same.

Example 6

Find the value of $(2.04 \times 10^6) + (4.70 \times 10^5)$.

$$\begin{aligned} (2.04 \times 10^6) + (4.70 \times 10^5) &= 2.04 \times 10^6 + 0.470 \times 10^6 \\ &= 2.51 \times 10^6 \end{aligned}$$

The value is 2.51×10^6 .

Significant figures in calculations

Addition and subtraction

The digits retained should only include those below and to the left of the least accurate quantity.

Example 7

Add 23.7 g, 0.53 g and 8.385 g.

23.7 The 7 in 23.7 is uncertain.
0.53 Thus retain only one decimal place.

$$\begin{array}{r} 8.385 \\ 23.7 \\ 0.53 \\ \hline 32.615 \end{array}$$

The total mass is 32.6 g.

Rounding off

When rounding off a number, use only the last digit to be discarded. When rounding off fives, the last digit retained should be even.

Example 8

Round the following to two significant figures: 7.25, 7.35, 7.251 and 7.249.

The values are

7.25 becomes 7.2

7.35 becomes 7.4

7.251 becomes 7.3

7.249 becomes 7.2

Multiplication and division

The number of significant figures in the solution should be equal to the least number of significant figures in any of the data.

Example 9

Calculate the volume of a rectangular block 2.35 cm by 128.3 cm by 1.8 cm.

$$\begin{aligned} l &= 128.3 \text{ cm} & V &= l \times b \times h \\ &= 1.283 \text{ m} & V &= (1.283 \text{ m}) \times (2.35 \times 10^{-2} \text{ m}) \times (1.8 \times 10^{-2} \text{ m}) \\ b &= 2.35 \text{ cm} & V &= 5.43 \times 10^{-4} \text{ m}^3 \\ &= 2.35 \times 10^{-2} \text{ m} & V &= 5.4 \times 10^{-4} \text{ m}^3 \text{ (two sig. fig.)} \\ h &= 1.8 \text{ cm} \\ &= 1.8 \times 10^{-2} \text{ m} \end{aligned}$$

The volume is $5.4 \times 10^{-4} \text{ m}^3$.

Example 10

Find the density of a sphere of radius 0.10 m and mass 12 kg.

$$\begin{aligned} r &= 0.10 \text{ m} & V &= \frac{4}{3}\pi r^3 = \frac{4}{3} \times \pi (10^{-1})^3 = 4.19 \times 10^{-3} \text{ m}^3 \\ m &= 12 \text{ kg} \\ \rho &= ? & \rho &= \frac{m}{V} = \frac{12}{4.19 \times 10^{-3}} = 2.9 \times 10^3 \text{ kg m}^{-3} \end{aligned}$$

The density is $2.9 \times 10^3 \text{ kg m}^{-3}$.

Set 2

- How many significant figures in the following:

(a) 125	(b) 0.013	(c) 1007
(d) 25.10	(e) 3700	(f) 0.003
- Convert the numbers in question 1 to standard notation.
- Calculate the area of a square of side 3.2 m.
- Find the volume in m^3 of a rectangular block 32 cm by 1.35 cm by 0.1 cm.
- Add the following lengths: 2.35 cm, 6.72 m, 14.2 m.
- Round off the following to three significant figures.

(a) 23571	(b) 2001	(c) 2754
(d) 31256	(e) 0.31356	(f) 1245
- Evaluate to the correct number of significant figures:

(a) $(2.35 - 1.72) \times 1.94 \times 17.67$
(b) $(2.9 + 3.2 + 7.1) \div 0.134$
- Evaluate the following:

(a) 2.3×10^{-6} multiplied by 3.2×10^5
(b) 3.74×10^{-9} divided by 1.23×10^{-11}
(c) Add 2.75×10^{-6} and 3.42×10^{-5}

Answers

- | | | | | | | |
|---|-------------------------------|-------|-------|----------------------------------|-------|---------------------------|
| 1 | (a) 3 | (b) 2 | (c) 4 | (d) 4 | (e) 2 | (f) 1 |
| 2 | (a) 1.25×10^2 | | | (b) 1.3×10^{-2} | | (c) 1.007×10^3 |
| | (d) 2.510×10^1 | | | (e) 3.7×10^3 | | (f) 3×10^{-3} |
| 3 | $1.0 \times 10^1 \text{ m}^2$ | | | 4 $4 \times 10^{-6} \text{ m}^3$ | | 5 20.9 m |
| 6 | (a) 2.36×10^4 | | | (b) 2.00×10^3 | | (c) 2.75×10^3 |
| | (d) 3.13×10^4 | | | (e) 3.14×10^{-1} | | (f) 1.24×10^3 |
| 7 | (a) 22 | | | (b) 98.5 | | |
| 8 | (a) 7.4×10^{-1} | | | (b) 3.04×10^2 | | (c) 3.70×10^{-5} |